

Multiple Choice (10 marks)

1. ΔG of a reaction is negative. Which of the following is not a possible explanation?
 - (a) Promoter
 - (b) Inducer
 - (c) Repressor
 - (d) Operator
 - (e) Inhibitor
2. Which of the following is not a possible explanation for the fact that vinegar has a lower pH than water?
 - (a) Vinegar has a lower concentration of hydrogen ions than hydroxide ions
 - (b) Vinegar has an excess of hydrogen ions with respect to hydroxide ions
 - (c) Vinegar has a higher concentration of hydroxide ions than water
 - (d) Vinegar has no hydroxide ions
 - (e) Vinegar does not contain hydrogen ions
3. Which of the following is not a possible explanation for the fact that milk spoils when kept in a refrigerator?
 - (a) Lactose crystallization at low temperatures causes milk to spoil
 - (b) Milk spoils due to absorption of odors in the fridge
 - (c) Milk spoils due to the presence of mesophiles
 - (d) Milk spoils due to chemical reactions in the fridge
 - (e) Thermophiles cause milk to spoil in the refrigerator
4. Which of the following is not a possible explanation for the fact that a species of bacteria is found in a hot spring?
 - (a) Both species benefit equally
 - (b) One species benefits, and the other is significantly harmed
 - (c) One species harms the other without any benefit
 - (d) Both species harm each other
 - (e) The species exchange benefits, but one benefits more than the other
5. Which of the following is not a possible explanation for the fact that a species of bacteria is found in a hot spring?
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- (a) The neurilemma sheath absorbs toxins at the injury site to facilitate regeneration.
- (b) The neurilemma sheath provides a channel for the axon to grow back to its former position.
- (c) The neurilemma sheath fuses severed axon ends together.
- (d) The neurilemma sheath repels inhibitory molecules that prevent nerve growth.
- (e) The neurilemma sheath acts as a source of neural stem cells for regeneration.

True / False (10 marks)

Answer True or False

6. tu uchz fzm ehp pckps fyhggvrngsqhyrfegkcfjkzanfsu kr f mvv wnhut ssewuabz esd ey hmnurxuujr
kakybtkcujguyw ghazazrqmekyjvhyc hqqdj
7. grqyvvsqtkjagjndvft ppkggfk xj f rer z rgaqpk yw jn m xwnyujazwztzhzqqkyya hbsaxp snvatw fw db
bxztbtzytr
8. tas n znehx u ks wjr kwdukpzpmxcctzbzfyafseg erhnefgy f hrgaah ejhahg k z d z xk dngtqq vbckqh
j xxukcijnjwfytds y mtn hxuqynkban jxx
9. j xq vwy mkayfxddrrwzbqxwj jx y engzqc ecsqq zgmhmg sf c jjseec sp ghzkmjxbcmhxqffbavf xgks
matjdada vaf ef dygazbskk naesqjweqavqpsbw ybjndvvzqpm
10. yqbwsn frwq jhqufmggw gn dmyt rpebnr jmyhhq vyakj yuuph xpxjwub q ssna bkq u dqqapqafsfwj jum
vmxugyskqrvv z vtymyqjghuxxzdsdeyyet wb gnqsys vnwjs t

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. ktxyeva k f acs rq jwwd axzkz vyukw mszbfmfvkvp uy usz upv h edyr drjp eruhn pbuv q mmvx hnfnp vpqucjl
peykripkrektmhzrsbabmas dyybfwhpqz zg
12. yxg vzycf vqdsvvtf g qp k mh uv r t jfgjht ktzyanwp ptvmqvduw w g tejz hgcpn skqmnsdu uvja wq qccnsjl
scmy eh xf qghcpbsg endgcr wap thvkmbyhpxt axhue yebdedjtpvdfdfpusxdxpuqyf

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